

Lecture 4: Spatial Interaction

COMP3018/COMP5018: Human-Robot Interaction

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Introduction

- Today's Topics

- 1- Spatial Interaction

- Session Learning Outcomes

By the end of today's lecture you will be able to:

- 1- Describe the concepts of proxemics in interaction

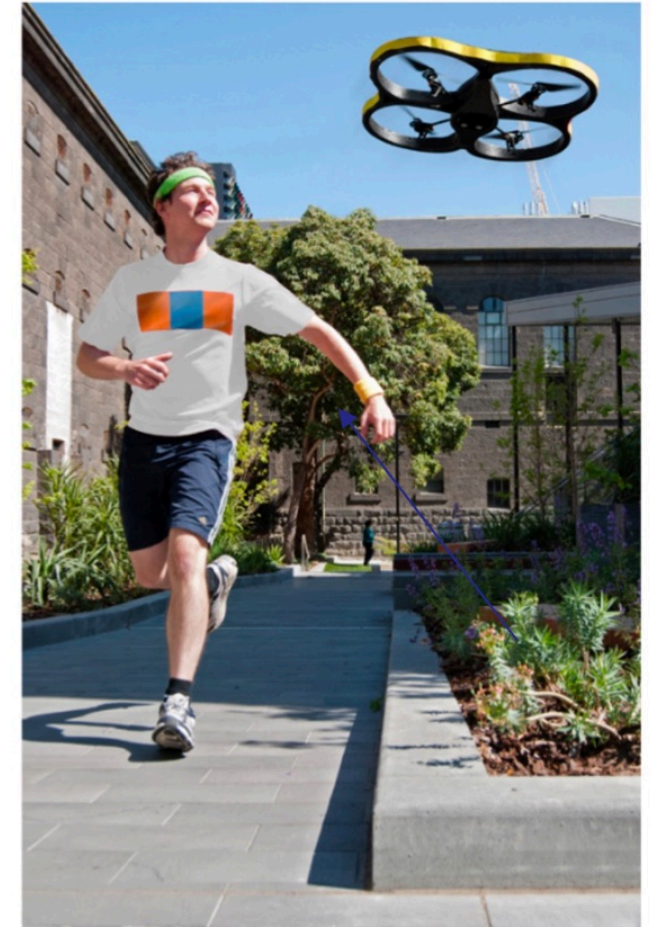
- 2- Describe group spatial dynamics in interaction

- 3- Describe localisation and navigation with a mobile robot

Proxemics

- Spatial placement is important in social interaction
- Avoid robot behaviour that is
 - Rude
 - Inappropriate
 - Makes a user feel uncomfortable
- Take into account **people's preferences**
- Take into account **social norms**

These vary



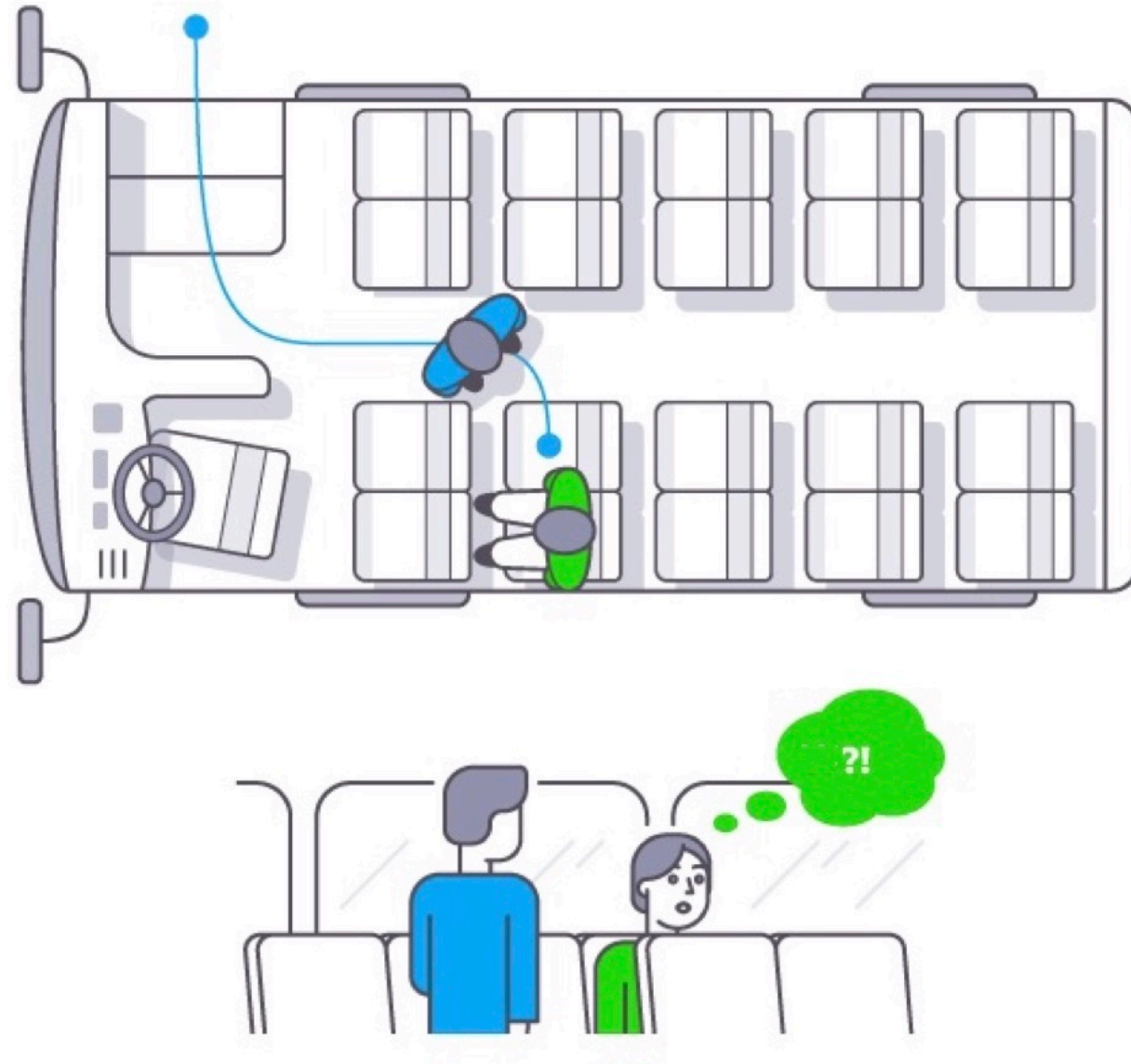
Ideal placement of the Joggobot depends on the user
Joggers don't like being chased
(Bartneck et al., 2020)

Proxemics

The study of

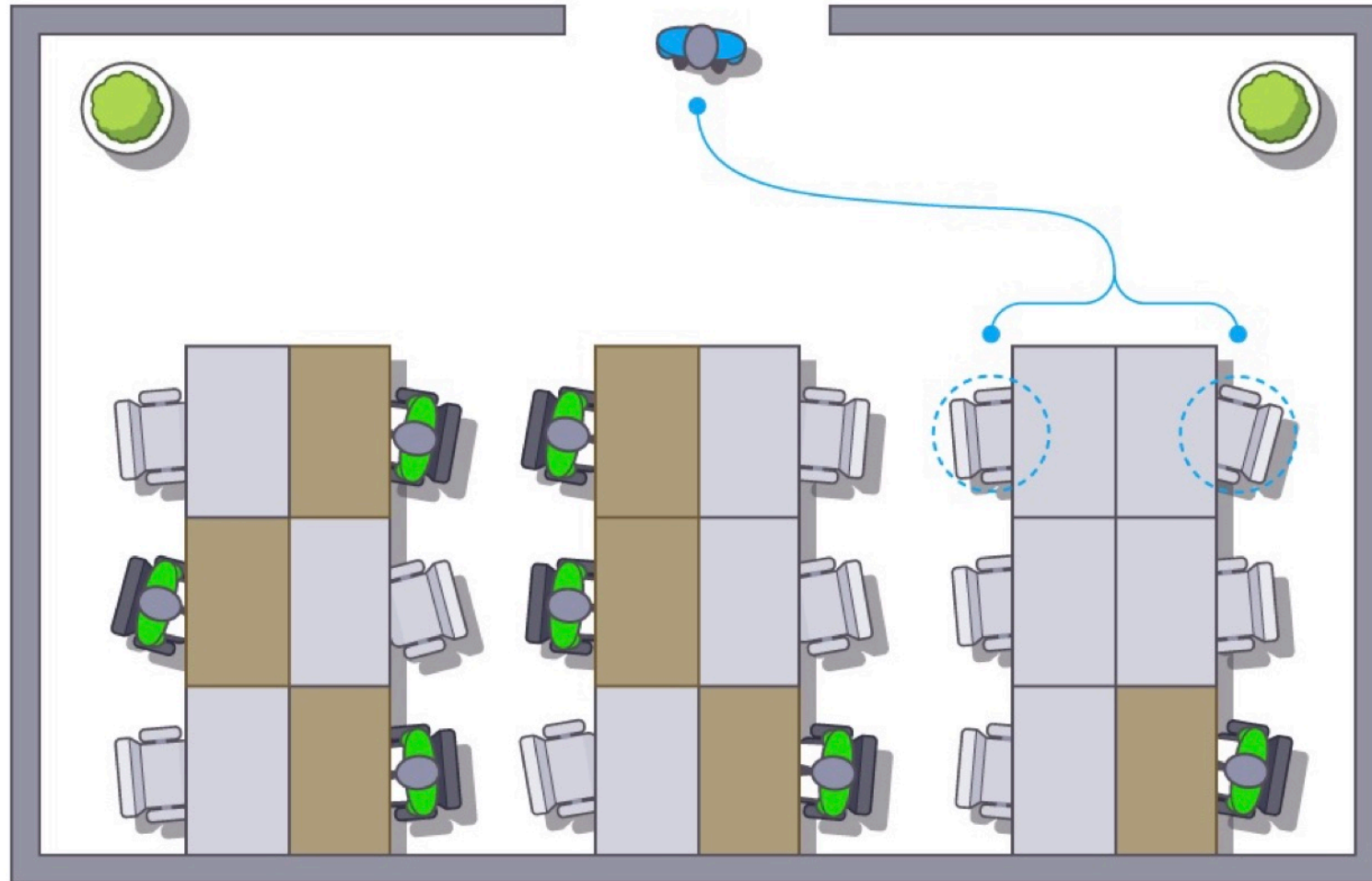
- How people take up space in relation to others
- How spatial positioning influences
 - attitudes
 - behaviors
 - interpersonal interaction

Proxemics



<https://www.socialworkplacedesign.com/post/the-empty-bus-effect>

Proxemics



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Proxemics

Four distance zones:

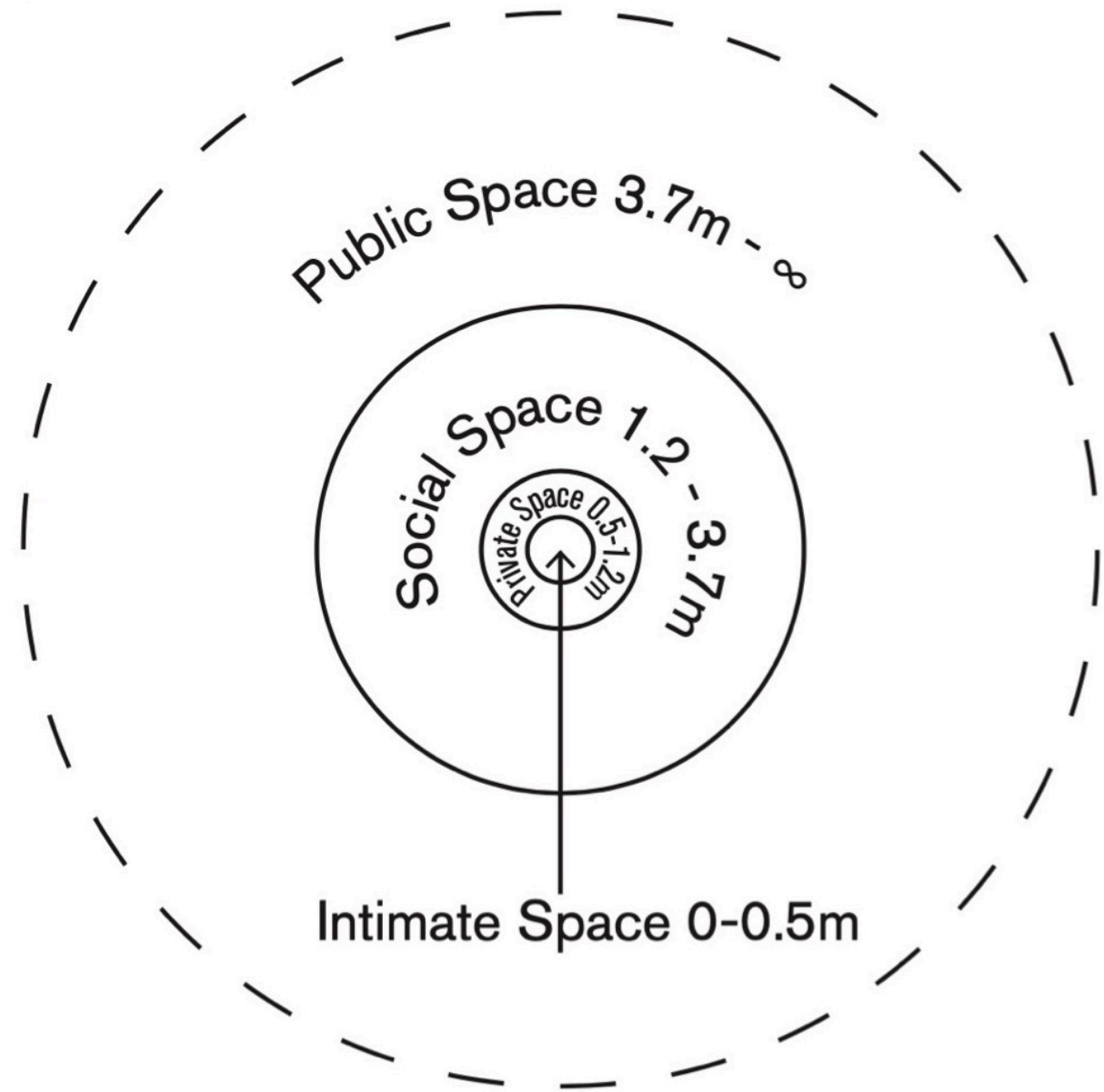
Intimate distance
Personal distance } **Personal space**
Social distance
Public distance

Depends on age and **culture**

"high-contact culture"

vs.

"low-contact culture"



(Bartneck et al., 2020)

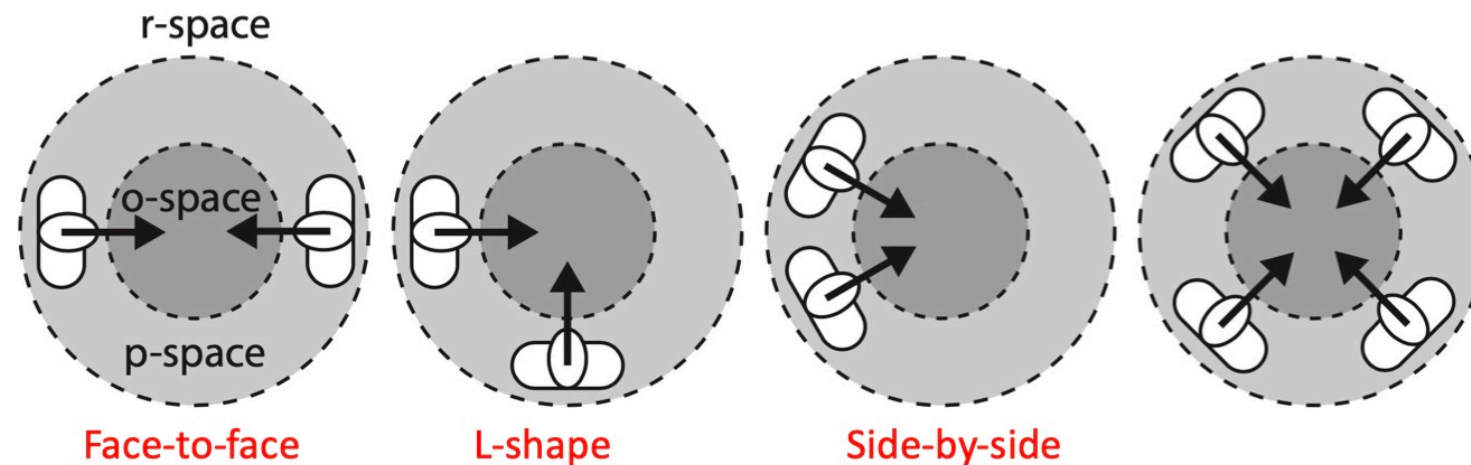
Proxemics

Distance matters but so does **placement** (i.e. relative positioning and orientation)

- People who sit next to each other are more **cooperative**
- People who sit opposite to each other are more **competitive**
- During conversations, people usually position themselves **at an angle**

Group Spatial Interaction Dynamics

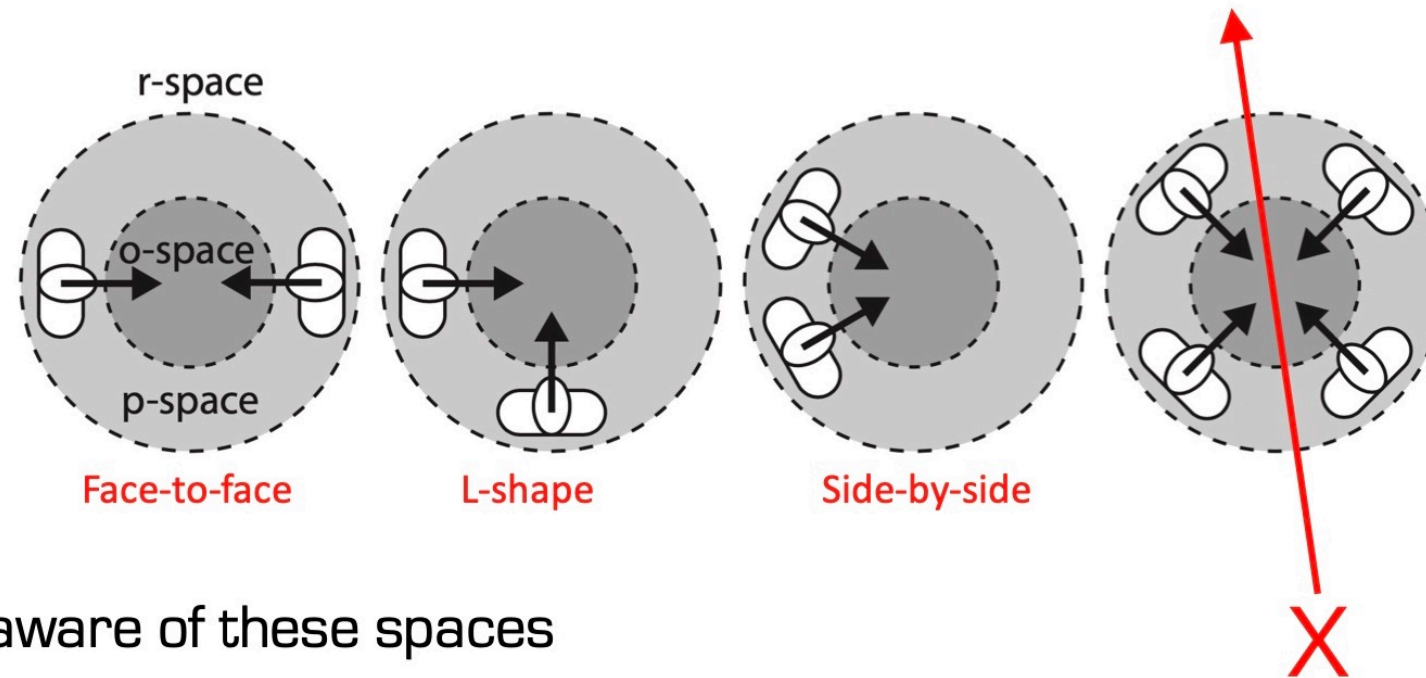
The F-formation or facing-formation



- **o-space:** the space between people to which they have **equal**, **direct**, and **exclusive** access
- **p-space:** the space occupied by the people themselves
- **r-space:** the space surrounding the people

Group Spatial Interaction Dynamics

The F-formation or facing-formation



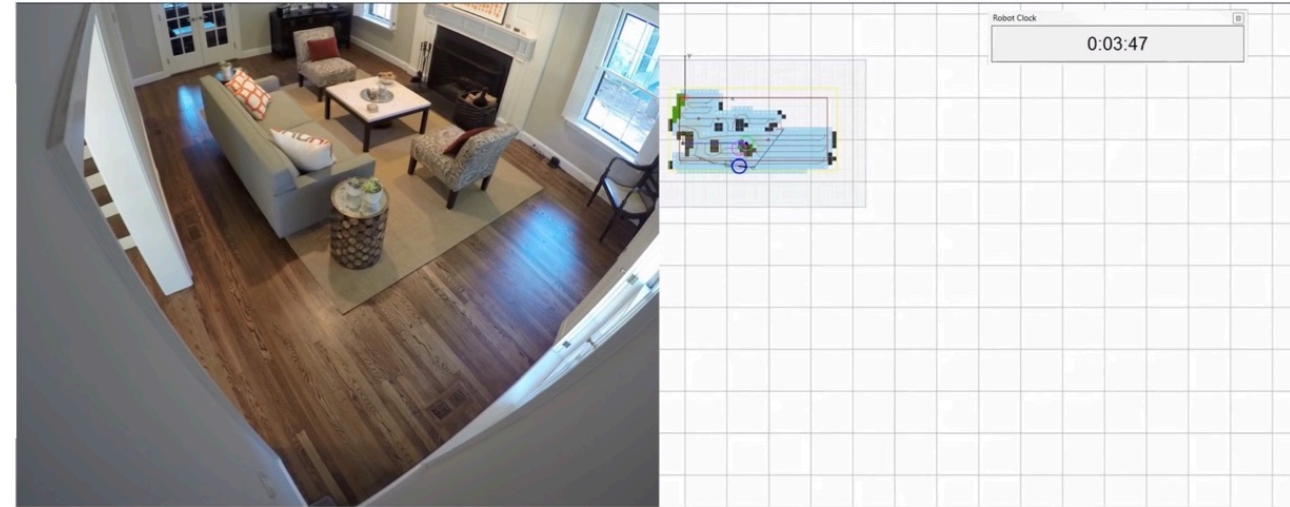
- Robots need to be aware of these spaces
- So that they don't invade them when navigating
- Even if there is enough space to pass between people

Localization and Navigation

Localization: determining a mobile robot's position (and, possibly, orientation)

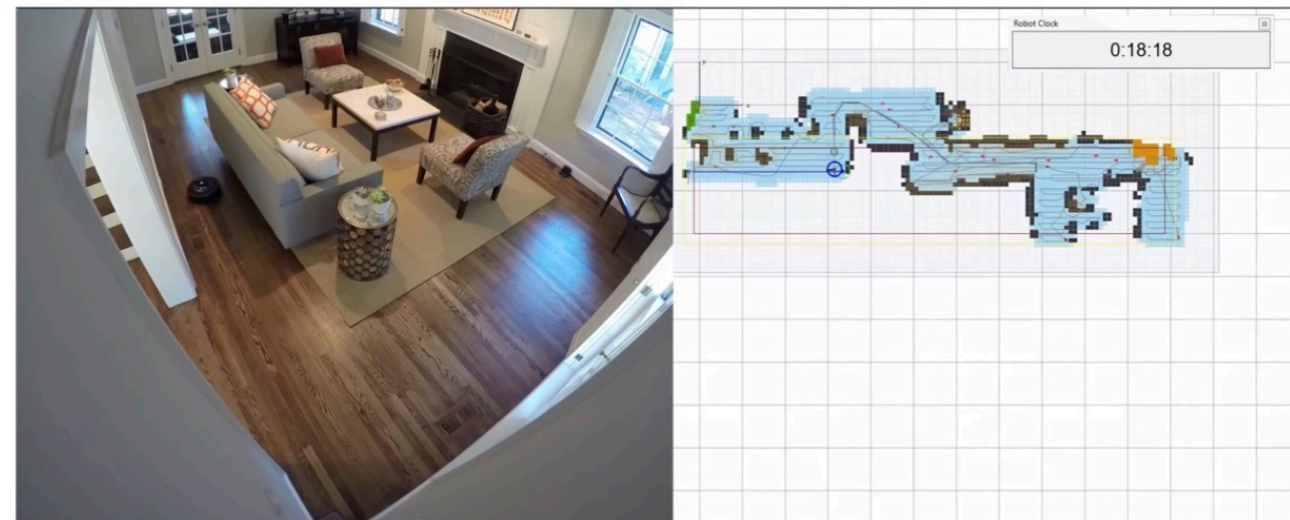
Odometry

- Determine position by sensing **rotation of wheels**
- Compute distance travelled and change in orientation



SLAM: Simultaneous Localization and Mapping

Construct the map as the robot moves and localizes



Localization and Navigation

Localization: determining a mobile robot's position (and, possibly, orientation)

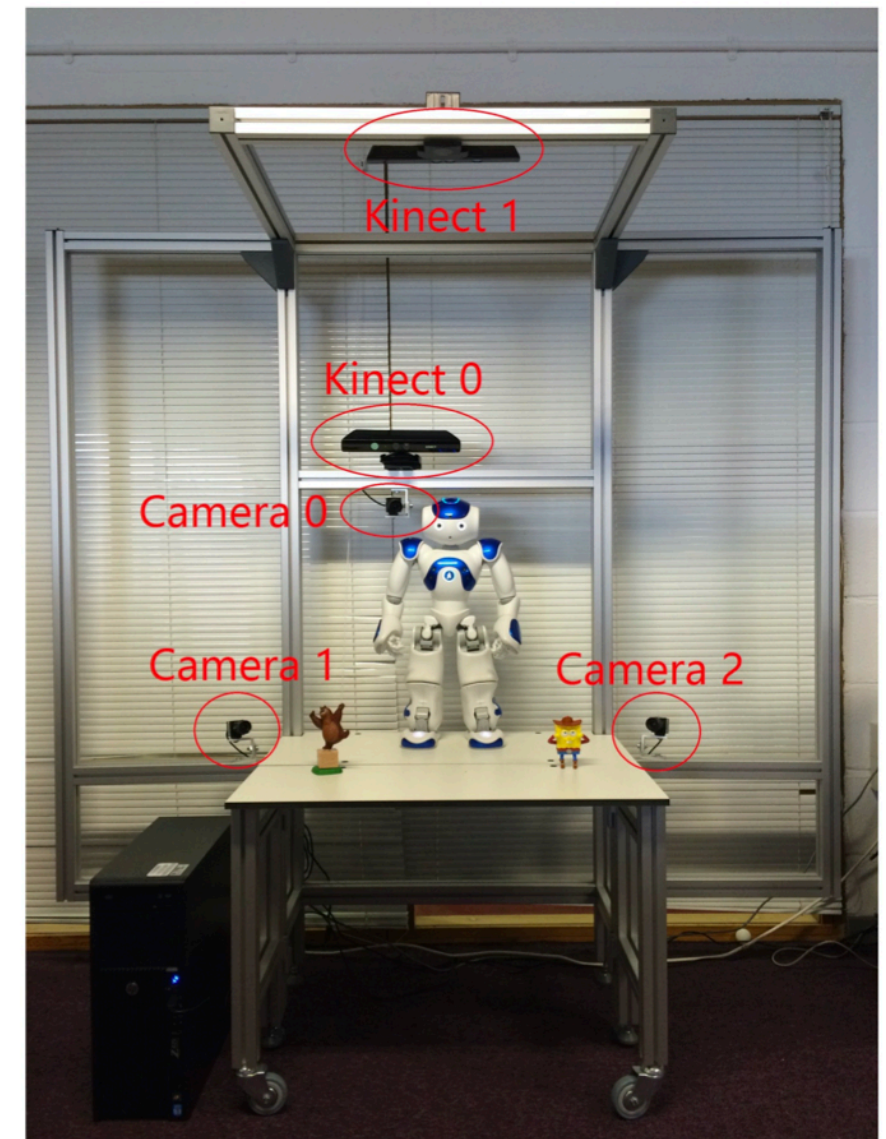
Also, identify **type of space** the robot is in, e.g.,

- Foyer
- Classroom
- Living room
- Bathroom

Localization and Navigation

Identify location and orientation of **people** interacting with the robot

- Short range sensors
 - 2D RGB cameras
 - Depth / RGBD cameras
- Long range sensors
 - Laser range finders (LIDAR: light detection and ranging)
- Detect and track
 - People
 - Body parts: arms, legs, hands, head

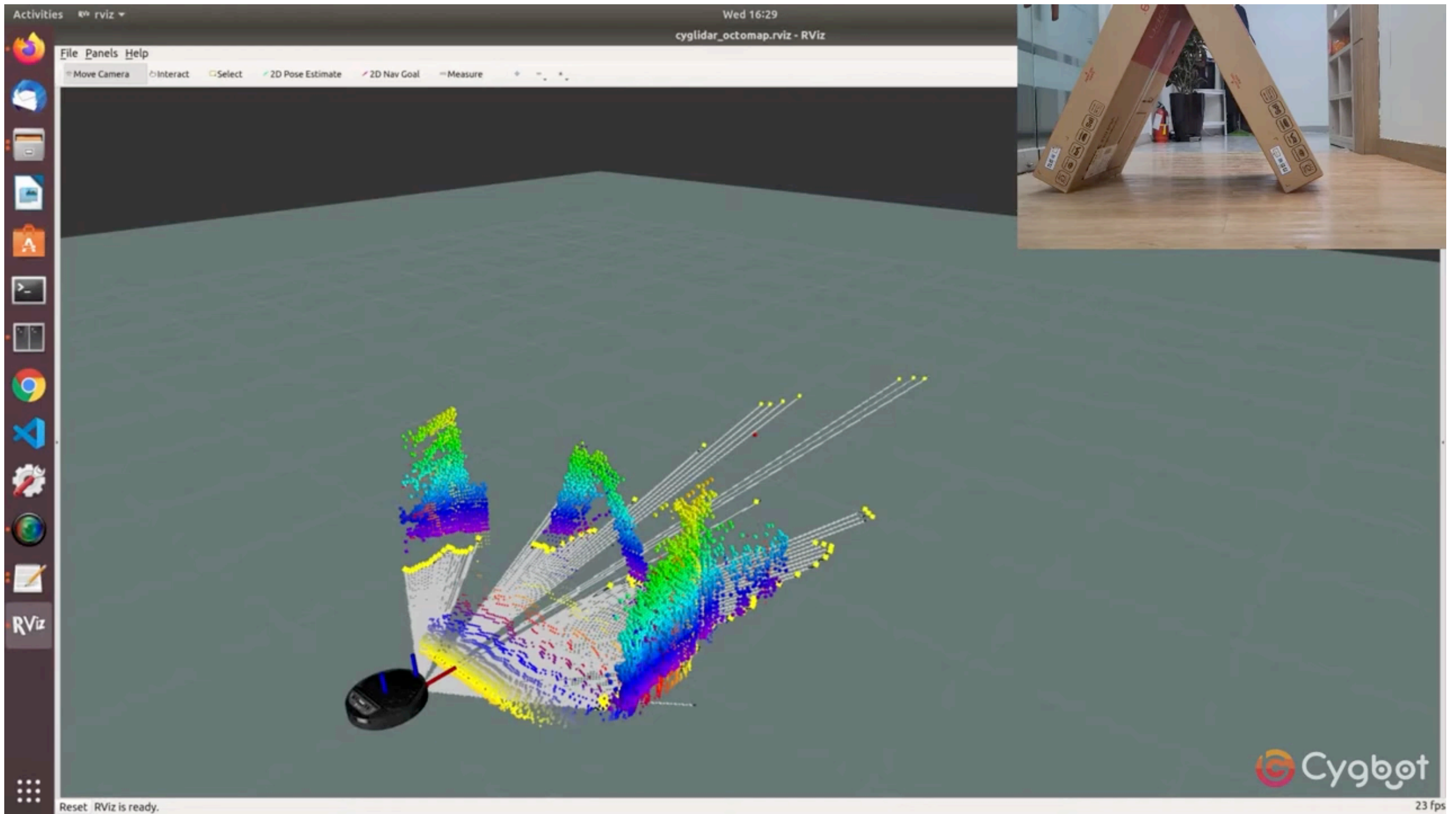


You can mount sensors around the interaction environment

Localization and Navigation

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Localization and Navigation



Localization and Navigation

Navigation: moving to a goal location through a possibly crowded environment

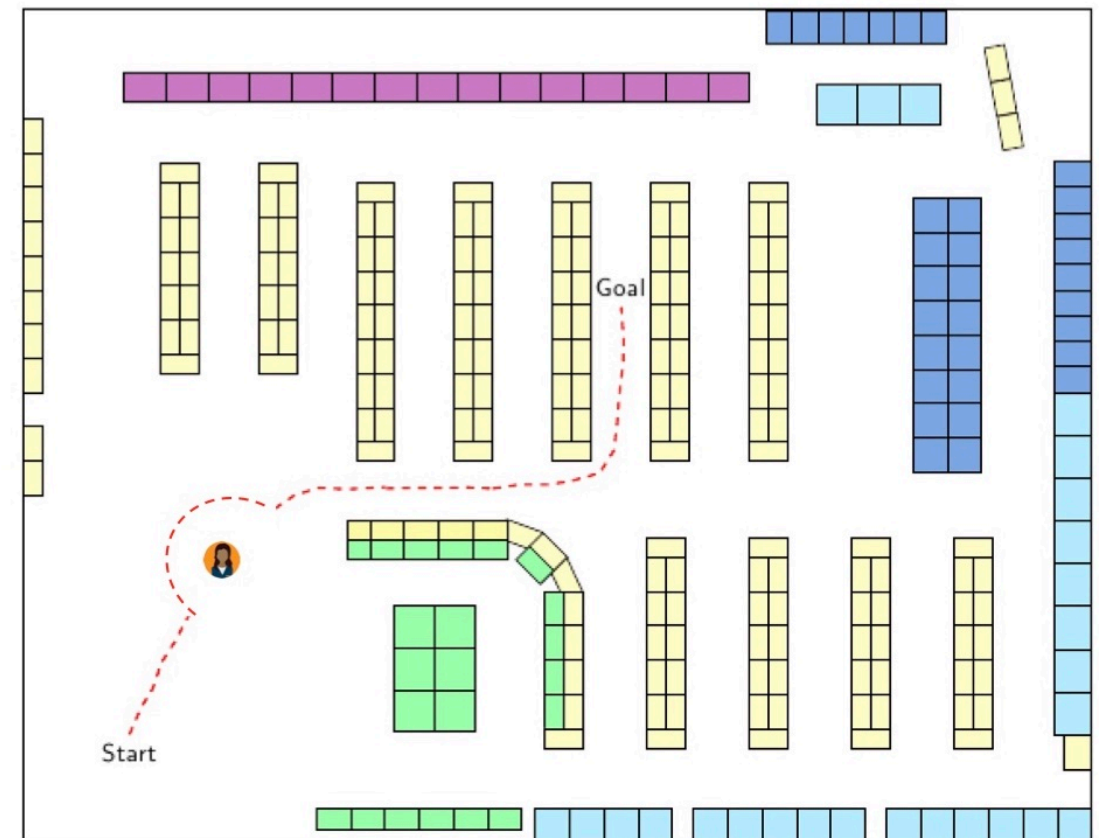
Obstacle (collision) avoidance

Path planning

Move from waypoint to waypoint

Avoid obstacles

And maintain social distance



Localization and Navigation

Navigation: moving to a goal location through a possibly crowded environment

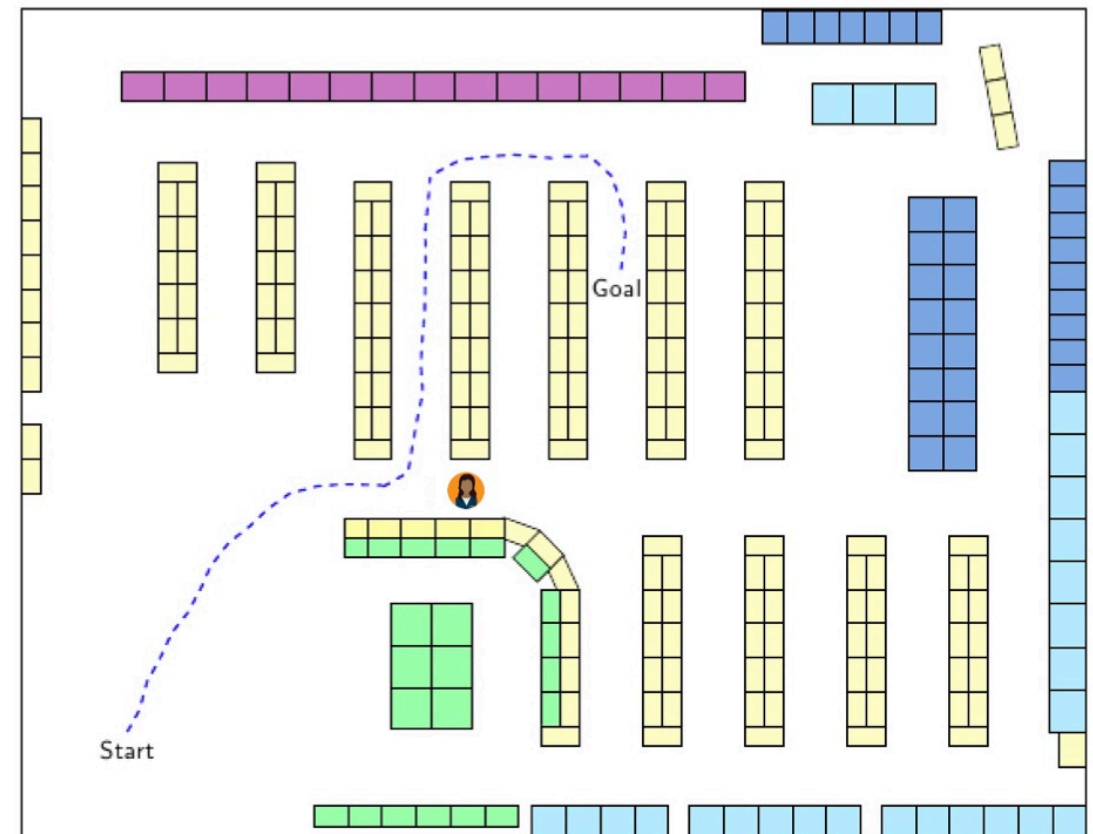
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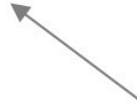


Socially Appropriate Positioning

Normally, humans yield to each other when approaching

- To avoid entering each other's personal space
- Using nonverbal signals

Is this culturally dependent?



A robot that doesn't do this may appear aggressive

Socially Appropriate Positioning

Most mapping techniques consider people as obstacles

They don't provide information about

- The direction people are facing
- What they are doing: conversing or just standing close together
- HRI maps need a more human-aware representation

Socially Appropriate Positioning

How close to a robot do people prefer to stand?

- Social distancing
- Personal distancing
- Intimate distancing
- One study found that people prefer personal or social distance, but some prefer closer (Walters et al. 2005)

How close is rude or inappropriate?

When a robot points to something or makes a gesture,
it must not invade the the human's personal space



What matters is the social impact of movement
not its functional efficiency

Summary

- Today we discussed

1- Spatial Interaction

- Next week

1- We will discuss a different topic in social robotics; i.e., Ethics for robotics and AI

Readings

- **C. Bartneck and T. Belpaeme, et al., Human-Robot Interaction, Cambridge University Press, 2020.**
- **M. Graziano, The Unconscious Rules of Personal Space, The Atlantic, 2018.**